

COMP 304 OPERATING SYSTEMS

Spring 2026

1 Course Information

Instructor:	Didem Unat, dunat@KU.EDU.TR
KU Credits:	3.00
ECTS Credits:	6.00
Prerequisite(s):	Prerequisite: COMP 201 or consent of the instructor
Class Location & Meeting Times:	SOS B08 - Monday, Wednesday 11:30-12:40
PS (Yes/No):	Yes
DS (Yes/No):	No
Lab (Yes/No):	No
Language of Instruction:	English
Office Hours:	Monday 10:30-11:30

2 Course Description

Introduction to operating systems concepts, process management, memory management, virtual memory, input-output and device management, file systems, job scheduling, threads, process synchronization, deadlocks, interrupt structures, case studies of operating systems.

3 Course Overview

An operating system is an essential part of any computer system. The purpose of this course is providing a clear understanding of the concepts that underlie operating systems. Fundamental concepts and algorithms that will be covered are based on those used in existing commercial operating systems. The aim is to present these topics in a general setting that is not tied to one particular operating system. Throughout the course, examples that pertain to the most popular operating systems such as Linux and Windows will be studied as well.

4 Course Learning Outcomes (CLOs)

CLO #	Upon successful completion of this course, students will be able to ...
1	Fundamental concepts in operating systems
2	Develop and understand concurrent programs and synchronization
3	Evaluate the efficiency aspect of using system resources and resource management
4	Understand relationships between operating system and computer architecture
5	Learn necessary concepts for memory management and file system management

5 Assessment Methods

Method	Description	Weight %
Projects	Three Projects (Implementations and Their Demos)	22.50
Project Exams	Project 1 and 2 Related Exam held together with Midterm	12.50
	Project 3 Related Exam held together with Final	10.00
Midterm Exam	Written Exam	20.00
Final Exam	Written Exam	30.00
Attendance	Participation and Attendance (Lecture and PS)	5.00
Total:		100.00

6 Instructional Material and Learning Resources

- Operating System Concepts, 10th ed. (ISBN: 1118129385)
 - Author: A. Silberschatz, Galvin, Gagne
 - Publisher: Wiley
 - Material Type: Textbook
 - Material Status: Required
- Linux Kernel Development, 3rd ed. (ISBN: 0672329468)
 - Author: Robert Love
 - Publisher: Addison-Wesley Professional
 - Material Type: Textbook
 - Material Status: Recommended

7 Course Schedule

Meeting Times	Subject
WEEK 1	Introduction, operating systems concepts and strategies, operating system structure organization, multiprogramming, brief history of operating systems, system calls, user and kernel modes, system boot
WEEK 2-3	Process Management: Process concept, process creation and termination, Unix processes, process tree, operations on processes, inter-process communication, cooperating processes, producer-consumer shared-memory solution
WEEK 4	CPU Scheduling: concepts, criteria, CPU scheduling algorithms, algorithm evaluation and performance, real-time CPU scheduling, Linux CPU scheduler
WEEK 5-6	Process Synchronization: race conditions, critical section problem and their solutions, semaphores, monitors, locks, busy waiting vs blocking, deadlock and starvation classical problems, condition variables, synchronization examples
WEEK 7	Thread Management: threads, multithreading models, POSIX Threads API, case studies of operating systems
WEEK 8	Deadlocks: system model, characterization, deadlock prevention, deadlock avoidance, deadlock detection and recovery
WEEK 9	Memory Management: memory allocation, internal and external fragmentation, swapping, paging, page table structure, segmentation, operating system examples
WEEK 10	Virtual Memory: demand paging, page fault, page replacement algorithms, allocation of frames, thrashing, working set model, operating system examples
WEEK 11	I/O Systems: mass storage structure, disk scheduling, file systems, file system implementation, protection and security, case studies
WEEK 12	File system interface, implementation and internals
WEEK 13	Virtualization, Containers, Docker
WEEK 14	Case studies: History of Linux and Windows

8 Student Code of Conduct and Academic Grievance Procedure

- [Student Code of Conduct](#)
- [Statement on Academic Honesty with Emphasis on Plagiarism](#)
- [Academic Grievance Procedure](#)

9 Course Policies

- Class Participation: Attendance will be taken randomly; participation in interactive discussions (i.e. Socratic or any other) during the class/online meetings might be used as part of student participation evaluation; participation will be evaluated based on the correctness of answers on discussion and not based on sole attendance.
- Project, Midterm and Final Exams: All exams are closed-book and will be conducted face-to-face.
- There will be a single make-up exam at the end of the semester for students who miss the midterm or the final (or both) due to health reasons. An approved medical report is required. The make-up exam may not follow the same format as the original exam(s) and

will be cumulative. If a student misses both the midterm and the final, the make-up exam will count toward the combined weight of both exams.

- Projects: Projects may be completed individually or in groups of two. Each group member is expected to contribute equally to all aspects of the project. Projects will be evaluated based on the implementation and an in-person demo. In addition, there will be a project exam during the midterm and another during the final exam, both of which assess your learning from the project. Therefore, students are expected to complete the projects on their own.
- Students who attend the midterm or final exam are also required to take the corresponding project exam. Although the grade contributions of the midterm/final exam and project exam are different, failure to attend either component will result in a zero score for that component. Therefore, students cannot take the midterm/final without taking the project exam, or vice versa.
- Course letter grading can be based on curve.
- All deadlines are final. Late project submissions will not be evaluated.